Lecture Notes

Lecture 12

Tensors and the Tensor Product

Lecture 11 introduced the indices; this lecture builds the objects they live on. The *tensor product* $V \otimes W$ is the vector space whose elements combine a vector from V with one from W multiplicatively, and tensors of higher type unify vectors, covectors, operators, and bilinear forms into one framework. Its matrix incarnation is the *Kronecker product*, and its physical meaning is decisive: finite composite pure-state vectors use the tensor product of the parts, and a nonzero pure state is *entangled* precisely when it fails to factor. This closes Part II of the course.

Remark 12.1 (Algebraic and quantum scope). This lecture's core uses finite-dimensional algebraic tensor products. General composite Hilbert spaces use the completed Hilbert tensor product $\mathcal{H}_A \mathbin{\widehat{\otimes}} \mathcal{H}_B$. The non-factorization criterion below is for nonzero pure states; a mixed state is separable when it is a convex combination of product density operators.

Learning objectives

By the end of this lecture you will be able to:

- Construct $V \otimes W$ via its basis and universal property; spot non-factoring tensors.
- Compute Kronecker products and use $(A \otimes B)(v \otimes w) = (Av) \otimes (Bw)$.
- Read a type-(p, q) tensor as a multilinear map and contract indices.
- Distinguish product from entangled nonzero pure states in finite composite quantum systems.

1 The tensor product

A bilinear map $V \times W \longrightarrow U$ is awkward—it is not linear. The tensor product repackages it as an ordinary linear map out of a single space.

Definition 12.2 (Tensor product). The *tensor product* $V \otimes W$ is the vector space with basis $\{e_i \otimes f_j\}$, where e_i is a basis of V and f_j a basis of W. Thus

$$\dim(V \otimes W) = (\dim V)(\dim W).$$

For $v = \sum_i v^i e_i$ and $w = \sum_j w^j f_j$, define

$$v \otimes w = \sum_{i,j} v^i w^j e_i \otimes f_j.$$

The *canonical map* $\iota \colon V \times W \longrightarrow V \otimes W$, $\iota(v, w) = v \otimes w$, is bilinear: $(v + v') \otimes w = v \otimes w + v' \otimes w$, $v \otimes (w + w') = v \otimes w + v \otimes w'$, and $(av) \otimes w = a(v \otimes w) = v \otimes (aw)$. Its value $v \otimes w$ is an *elementary* (or decomposable) tensor—an element, not itself a bilinear map.

A general element of $V \otimes W$ is a sum $\sum_{ij} \tau^{ij} e_i \otimes f_j$ of elementary tensors; crucially, most tensors are *not* themselves elementary—this is the algebraic seed of entanglement.

$e_1 \otimes f_1$	$e_1 \otimes f_2$	$e_1 \otimes f_3$
$e_2 \otimes f_1$	$e_2 \otimes f_2$	$e_2 \otimes f_3$

Figure 1: A basis of $V \otimes W$ for $\dim V = 2$, $\dim W = 3$: every pair (e_i, f_j) contributes one basis tensor, giving $\dim(V \otimes W) = 2 \cdot 3 = 6$.

Example 12.3 (A tensor that does not factor). In $\mathbb{C}^2 \otimes \mathbb{C}^2$ the tensor $e_1 \otimes e_1 + e_2 \otimes e_2$ is not elementary: if it equalled $v \otimes w$ with $v = (a, b)$, $w = (c, d)$, its components would be the rank-one array $\left(\begin{smallmatrix} ac & ad \\ bc & bd \end{smallmatrix}\right)$, whereas the given tensor has components $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, of rank two. An algebraic tensor is decomposable exactly when its component matrix has rank at most one: the zero tensor is $0 \otimes w$ and has rank zero, while every nonzero decomposable tensor has rank one. ◁

Example 12.4 (Functions of two variables). The tensor product of polynomial spaces realizes products of functions: $\mathcal{P}_m(x) \otimes \mathcal{P}_n(y)$ is spanned by $x^i \otimes y^j$, identified with the monomials $x^i y^j$. A general element—a polynomial in x and y—is a sum of products $f(x)g(y)$, and is a single product exactly when it factors. ◁

Proposition 12.5 (Universal property). For every bilinear map $B \colon V \times W \to X$, there is a unique linear map $\widetilde{B} \colon V \otimes W \to X$ such that $\widetilde{B}(v \otimes w) = B(v, w)$ for all v, w. Thus $B = \widetilde{B} \circ \iota$.

Proof. On basis tensors set $\widetilde{B}(e_i \otimes f_j) = B(e_i, f_j)$ and extend linearly. For arbitrary $v = \sum_i v^i e_i$ and $w = \sum_j w^j f_j$, bilinearity gives

$$\widetilde{B}(v \otimes w) = \sum_{i,j} v^i w^j B(e_i, f_j) = B(v, w).$$

Any linear map with this property has the same value on every basis tensor $e_i \otimes f_j$, so it is unique. This factorization is the basis-independent characterization of the tensor product. ■

Remark 12.6 (Many factors). There is a canonical associator isomorphism $U \otimes (V \otimes W) \cong (U \otimes V) \otimes W$ sending $u \otimes (v \otimes w)$ to $(u \otimes v) \otimes w$, so multi-factor products $V_1 \otimes \cdots \otimes V_p$ may suppress parentheses; their basis is the products $e_{i_1} \otimes \cdots \otimes e_{i_p}$ and their dimension is the product of the dimensions. This is the home of (p, q) tensors and of many-body quantum states.

2 The Kronecker product

In coordinates the tensor product is the *Kronecker product*: stacking the basis $e_i \otimes f_j$ lexicographically, the column of $v \otimes w$ is the Kronecker product of the columns of v and w, and an operator $A \otimes B$ has the block form

$$A \otimes B = \begin{pmatrix} A_{11}B & \cdots & A_{1n}B \\ \vdots & & \vdots \\ A_{n1}B & \cdots & A_{nn}B \end{pmatrix},$$

each entry A_{ij} of A replaced by the block $A_{ij}B$.

Proposition 12.7 (Properties of $A \otimes B$). The Kronecker product satisfies

$$(A \otimes B)(v \otimes w) = (Av) \otimes (Bw), \qquad (A \otimes B)(C \otimes D) = (AC) \otimes (BD),$$

together with $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$, $\operatorname{tr}(A \otimes B) = (\operatorname{tr} A)(\operatorname{tr} B)$, and if $Av = \lambda v$, $Bw = \mu w$ then $(A \otimes B)(v \otimes w) = \lambda\mu\,(v \otimes w)$.

Proof. By linearity it suffices to check the identities on elementary tensors. From the block form, $(A\otimes B)(e_i \otimes f_j) = \sum_{a,b} A_{ai}B_{bj}\, e_a \otimes f_b = (Ae_i)\otimes(Bf_j)$, so $(A\otimes B)(v\otimes w) = (Av)\otimes(Bw)$. The composition and eigenvalue identities follow at once: $(A\otimes B)(C\otimes D)(v\otimes w) = (ACv)\otimes(BDw) = \big((AC)\otimes(BD)\big)(v\otimes w)$, and $(A\otimes B)(v\otimes w) = (\lambda v)\otimes(\mu w) = \lambda\mu\,(v\otimes w)$ when $Av = \lambda v$, $Bw = \mu w$. For the adjoint, using $\langle v\otimes w, v'\otimes w'\rangle = \langle v, v'\rangle\langle w, w'\rangle$ and $\langle Av, x\rangle = \langle v, A^\dagger x\rangle$,

$$\begin{aligned}\langle (A\otimes B)(v\otimes w), x\otimes y\rangle &= \langle Av, x\rangle\langle Bw, y\rangle = \langle v, A^\dagger x\rangle\langle w, B^\dagger y\rangle \\ &= \langle v\otimes w, (A^\dagger \otimes B^\dagger)(x\otimes y)\rangle,\end{aligned}$$

so $(A\otimes B)^\dagger = A^\dagger \otimes B^\dagger$. Finally, summing the diagonal of the block form, $\operatorname{tr}(A\otimes B) = \sum_{a,b} A_{aa}B_{bb} = (\operatorname{tr} A)(\operatorname{tr} B)$. ■

For finite square matrices over $\mathbb{C}$ (or after complexifying real matrices), the spectrum of $A\otimes B$ is $\{\lambda_i\mu_j\}$ with algebraic multiplicity. Over $\mathbb{R}$ this need not describe the real point spectrum: the real quarter-turn R has no real eigenvalue, while $R\otimes R$ has real eigenvalues $1, 1, -1, -1$. Over either field, tensor products of diagonalizable operators are diagonalizable when each factor is diagonalizable over that field. Tensor products of normal operators are normal on finite inner-product spaces.

Example 12.8 (A Kronecker product of Pauli matrices). With $\sigma_x = \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$,

$$\sigma_x \otimes \sigma_x = \begin{pmatrix}0 & \sigma_x\\ \sigma_x & 0\end{pmatrix} = \begin{pmatrix}0&0&0&1\\0&0&1&0\\0&1&0&0\\1&0&0&0\end{pmatrix},$$

the operator that flips both qubits. Its eigenvalues are the products $(\pm1)(\pm1)$, namely $+1$ (twice, on $|+\rangle\otimes|+\rangle$ and $|-\rangle\otimes|-\rangle$) and -1 (twice, on $|+\rangle\otimes|-\rangle$ and $|-\rangle\otimes|+\rangle$)—consistent with $\operatorname{tr}(\sigma_x\otimes\sigma_x) = (\operatorname{tr}\sigma_x)^2 = 0$. ◁

Example 12.9 (Kronecker product of vectors). For $v = (1,2)$ and $w = (3,4)$,

$$v\otimes w = (1\cdot w,\ 2\cdot w) = (3,4,6,8),$$

the four products v_iw_j in lexicographic order. The two-qubit state $|0\rangle\otimes|1\rangle$ is likewise the column $(0,1,0,0) = |01\rangle$. ◁

Example 12.10 (Acting factorwise). With $A = \sigma_x$, $B = \sigma_z$ and $|00\rangle = |0\rangle\otimes|0\rangle$, $(\sigma_x\otimes\sigma_z)|00\rangle = (\sigma_x|0\rangle)\otimes(\sigma_z|0\rangle) = |1\rangle\otimes|0\rangle = |10\rangle$: each factor acts on its own slot, the content of $(A\otimes B)(v\otimes w) = (Av)\otimes(Bw)$. ◁

Example 12.11 (A single-qubit gate on two qubits). To apply a Hadamard H to the first qubit only, use $H\otimes I$: it acts as H on the first slot and leaves the second alone, $(H\otimes I)|0x\rangle = |+\rangle\otimes|x\rangle$. Local gates are tensor products with identities, and they cannot create entanglement. ◁

Example 12.12 (A two-qubit gate matrix). The Hadamard on both qubits is

$$H \otimes H = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix},$$

each scalar entry $\frac{1}{\sqrt{2}}H_{ij}$ scaling the block $\frac{1}{\sqrt{2}}H$. Applied to $|00\rangle$ it produces the uniform superposition $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$. ◁

$$\begin{array}{|c|c|} \hline A_{11}B & A_{12}B \\ \hline A_{21}B & A_{22}B \\ \hline \end{array}$$

Figure 2: The Kronecker product $A \otimes B$ for a 2×2 matrix A: each scalar entry A_{ij} becomes the block $A_{ij}B$, building an $(nm) \times (nm)$ matrix from $n \times n$ and $m \times m$ pieces.

Example 12.13 (The mixed-product rule). The rule $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$ lets gates compose factorwise. For instance $(\sigma_x \otimes I)(\sigma_z \otimes I) = (\sigma_x\sigma_z) \otimes I = (-i\sigma_y) \otimes I$, so applying σ_z then σ_x to the first qubit is the single gate $-i\sigma_y \otimes I$. ◁

Remark 12.14 (The swap operator). The swap $|ij\rangle \mapsto |ji\rangle$ exchanges the two factors of $\mathcal{H} \otimes \mathcal{H}$. It acts as $+1$ on the symmetric tensors $S^2\mathcal{H}$ and -1 on the antisymmetric tensors $\Lambda^2\mathcal{H}$ (both defined in §4 below), so the $S^2 \oplus \Lambda^2$ split is exactly its eigenspace decomposition.

Example 12.15 (Single-qubit operators on two qubits). On two qubits $\sigma_z \otimes I$ and $I \otimes \sigma_z$ are the diagonal matrices

$$\sigma_z \otimes I = \mathrm{diag}(1, 1, -1, -1), \qquad I \otimes \sigma_z = \mathrm{diag}(1, -1, 1, -1),$$

measuring the first and second spin respectively. They commute, and their product is $\sigma_z \otimes \sigma_z = \mathrm{diag}(1, -1, -1, 1)$, the parity. ◁

Remark 12.16 (Trace, determinant, and size). For finite square matrices, $\det(A \otimes B) = (\det A)^m(\det B)^n$ for A of size n and B of size m. The identity follows by triangularizing over $\mathbb{C}$ and then, for real matrices, by the same polynomial identity; it does not rely on a false real-spectrum product formula. The dimensions check out: $A \otimes B$ is $nm \times nm$, matching $\dim(V \otimes W) = nm$.

Remark 12.17 (Inner product on a tensor product). If V, W are finite-dimensional inner-product spaces, $V \otimes W$ inherits the inner product $\langle v \otimes w, v' \otimes w' \rangle = \langle v, v' \rangle \langle w, w' \rangle$ and an orthonormal basis $\{e_i \otimes f_j\}$ whenever the chosen factor bases are orthonormal. Arbitrary chosen bases need not be orthonormal. In general Hilbert spaces the algebraic product is usually incomplete; the quantum composite is the completion $\mathcal{H}_A \widehat{\otimes} \mathcal{H}_B$. The finite results of Lectures 7–10 apply directly to finite-dimensional composites. Lecture 13 introduces the completeness and Hilbert-space machinery that explains why completion is needed; constructing completed tensor products and developing infinite-dimensional Schmidt theory are beyond this course.

3 Tensors as multilinear maps

Throughout this section V, W are finite-dimensional. A *tensor of type* (p, q) is a multilinear map taking p covectors and q vectors to a scalar; equivalently it is an element of $V^{\otimes p} \otimes (V^*)^{\otimes q}$, with components $\tau^{i_1\cdots i_p}{}_{j_1\cdots j_q}$ carrying p upper and q lower indices. The familiar objects are low-rank tensors: a vector is $(1,0)$, a covector $(0,1)$, a real bilinear form is $(0,2)$, and a linear operator is $(1,1)$.

Proposition 12.18 (Operators are tensors). There is a canonical isomorphism

$$\mathcal{L}(V,W) \cong W \otimes V^*, \qquad A^a{}_i \longmapsto A^a{}_i\, w_a \otimes e^i,$$

sending an operator to the tensor built from its matrix. The identity operator corresponds to $\delta^a{}_i\, w_a \otimes e^i$—the Kronecker delta is the identity tensor when $W = V$ and the same basis is used in both slots.

Proof. Define $\Psi\colon W \otimes V^* \to \mathcal{L}(V,W)$ on elementary tensors by $\Psi(w \otimes \varphi)(v) = \varphi(v)\, w$, extended linearly; this uses only the pairing of V^* with V, so it needs no basis. On the basis $\{w_a \otimes e^i\}$, $\Psi(w_a \otimes e^i)(e_j) = e^i(e_j)\, w_a = \delta^i_j\, w_a$, the elementary operator $E^a{}_i$ sending $e_i \mapsto w_a$ and $e_j \mapsto 0$ $(j \neq i)$. As $\{E^a{}_i\}$ is a basis of $\mathcal{L}(V,W)$, Ψ carries a basis to a basis and is an isomorphism, with $\dim = (\dim W)(\dim V)$. Its inverse sends $A = A^a{}_i$ to $A^a{}_i\, w_a \otimes e^i$, the correspondence of the statement; that tensor returns $v \mapsto A^a{}_i\, e^i(v)\, w_a = A^a{}_i v^i w_a = Av$. ■

For arbitrary algebraic spaces the same canonical map $W \otimes V^* \to \mathcal{L}(V,W)$ reaches the finite-rank operators, not generally every linear map; in particular, the identity on an infinite-dimensional V is not a finite tensor.

Example 12.19 (Outer products are decomposable operators). The rank-one operator $|a\rangle\langle b|$ corresponds to the elementary tensor $|a\rangle \otimes \langle b| \in V \otimes V^*$. A general operator is a sum of such outer products (its tensor is a sum of decomposables), and it is itself decomposable—rank at most one—exactly when it is a single outer product $|a\rangle\langle b|$. A nonzero such operator has rank one; the zero operator is decomposable with rank zero. ◁

Example 12.20 (A real metric as a $(0,2)$ tensor). Over $\mathbb{R}$, a metric or pseudo-metric is the bilinear $(0,2)$ tensor $g = g_{ij}\, e^i \otimes e^j$ and $g(u,v) = g_{ij} u^i v^j$; if it is nondegenerate it can lower and raise indices. Over $\mathbb{C}$, the course inner product is instead Hermitian sesquilinear: $\langle u, v\rangle = \overline{u^i}\, g_{ij} v^j$, and its Gram matrix transforms as $g' = P^\dagger g P$, not $P^\top g P$. ◁

Example 12.21 (An operator as components). The operator $T = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right)$ has $(1,1)$ components $T^1{}_1 = 2, T^1{}_2 = 1, T^2{}_1 = 0, T^2{}_2 = 3$, and as a tensor $T = 2\, e_1 \otimes e^1 + e_1 \otimes e^2 + 3\, e_2 \otimes e^2$. Its trace is the contraction $T^i{}_i = 2 + 3 = 5$. ◁

Example 12.22 (Trace via the identity tensor). Contracting the identity tensor $\delta^i{}_j$ with itself gives $\delta^i{}_i = \sum_i 1 = n = \dim V$, so $\operatorname{tr} I = \dim V$. The Kronecker delta is the identity operator, and contracted it is the dimension. ◁

The single operation that lowers tensor rank is *contraction*: summing over a matched upper and lower index, $\tau^i{}_i$, which for a $(1,1)$ tensor is exactly the trace. Under a change of basis each upper index transforms with P^{-1} and each lower index with P (Lecture 11), so a contracted pair is invariant.

Example 12.23 (Contraction is the trace). For an operator $T = T^i{}_j\, e_i \otimes e^j$, contracting the upper against the lower index gives $T^i{}_i = \sum_i T^i{}_i = \operatorname{tr} T$, basis-independent by Lecture 11. More generally, contracting one pair of a higher tensor lowers its type by $(1,1)$, turning, say, a $(1,2)$ tensor into a covector. ◁

Example 12.24 (A real bilinear transformation law). A $(1,1)$ tensor transforms as $T'^i{}_j = (P^{-1})^i{}_k T^k{}_l P^l{}_j$, a $(0,2)$ tensor as $g'_{ij} = P^k{}_i P^l{}_j\, g_{kl}$, and a $(2,0)$ tensor with two factors of P^{-1}. The rule is mechanical: one factor of P^{-1} per upper index, one of P per lower index. ◁

Example 12.25 (Bridge: bivectors and the familiar cross product). An antisymmetric $(2,0)$ tensor is a *bivector*. In oriented Euclidean $\mathbb{R}^3$, the metric and orientation determine a Hodge star, and the familiar cross product is $u \times v = \star(u \wedge v)$. Equal dimensions alone do not give a canonical identification $\Lambda^2 V \cong V$. ◁

4 Bridge: symmetric and antisymmetric tensors

Within $V \otimes V$ live two natural subspaces.

> **Definition 12.26 (Symmetric and antisymmetric tensors).** A tensor in $V \otimes V$ is *symmetric* if swapping its two slots leaves it unchanged and *antisymmetric* if swapping flips the sign. These form subspaces $S^2 V$ and $\Lambda^2 V$, and
>
> $$V \otimes V = S^2 V \oplus \Lambda^2 V.$$
>
> They are built from vectors by the symmetrized product $v \otimes w + w \otimes v$ and the wedge $v \wedge w = v \otimes w - w \otimes v$.

The wedge product is alternating ($v \wedge v = 0$) and feeds the theory of determinants and differential forms (Lecture 11). The split mirrors the decomposition of a square matrix into symmetric and antisymmetric parts. In the standard nonrelativistic bridge, these subspaces model bosons and fermions. Exchange symmetry applies to the *total* one-particle degrees of freedom, including spatial and internal factors, not to spin alone; this is not a complete spin–statistics treatment.

Example 12.27 (Splitting a tensor). Any $\tau = \sum \tau^{ij} e_i \otimes e_j$ splits as $\tau = \tau_+ + \tau_-$ with symmetric part $\tau_+^{ij} = \frac{1}{2}(\tau^{ij} + \tau^{ji})$ and antisymmetric part $\tau_-^{ij} = \frac{1}{2}(\tau^{ij} - \tau^{ji})$. For $e_1 \otimes e_2$ this gives $\frac{1}{2}(e_1 \otimes e_2 + e_2 \otimes e_1) + \frac{1}{2}\, e_1 \wedge e_2$—a symmetric piece plus a wedge. ◁

Example 12.28 (Symmetric tensors are polynomials). Symmetric tensors in $S^k V^*$ correspond to homogeneous degree-k polynomials in the coordinates, $e^i \otimes_S e^j \leftrightarrow x_i x_j$. Their dimension $\binom{n+k-1}{k}$ counts degree-k monomials in n variables: symmetric tensor algebra is polynomial algebra. ◁

Remark 12.29 (Identical particles). Quantum statistics is the symmetry of multi-particle tensors: n identical bosons live in the symmetric power $S^n \mathcal{H}$, n fermions in $\Lambda^n \mathcal{H}$. Since Λ^n vanishes once n exceeds $\dim \mathcal{H}$, no two fermions occupy the same complete one-particle state—the Pauli principle in this finite nonrelativistic model.

Remark 12.30 (Dimensions). For $\dim V = n$, $\dim S^2 V = \binom{n+1}{2}$ and $\dim \Lambda^2 V = \binom{n}{2}$, summing to $n^2 = \dim(V \otimes V)$. At the top, $\Lambda^n V$ is one-dimensional—its single basis element is a chosen volume element. The determinant is the alternating coefficient/map defined relative to that element.

Example 12.31 (The wedge and the determinant). In $\Lambda^2 \mathbb{R}^2$, $u \wedge v = (u_1 v_2 - u_2 v_1)\, e_1 \wedge e_2$: the lone coefficient is the 2×2 determinant. In general

$$v_1 \wedge \cdots \wedge v_n = \det[v_1 \ \cdots \ v_n]\, (e_1 \wedge \cdots \wedge e_n),$$

so the determinant is the component of the top wedge—volume as an antisymmetric tensor. ◁

Example 12.32 (Two-forms in three dimensions). $\Lambda^2 \mathbb{R}^3$ has dimension $\binom{3}{2} = 3$, with basis $e_1 \wedge e_2, e_2 \wedge e_3, e_3 \wedge e_1$—the same count as vectors in $\mathbb{R}^3$. After choosing an orientation and Euclidean metric, the Hodge star turns $u \wedge v$ into $u \times v$. Dimension equality alone is not canonical; moreover, normed bilinear vector cross products also have a seven-dimensional exception (from the imaginary octonions). ◁

5 Tensor products in quantum mechanics

The pure-state vector space of a finite composite system is the tensor product of the parts: $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$. For two qubits, $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$ with orthonormal basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ (writing $|ij\rangle = |i\rangle \otimes |j\rangle$).

Definition 12.33 (Product and entangled states). A nonzero pure state of $\mathcal{H}_A \otimes \mathcal{H}_B$ is a *product state* if a representing ket factors as $|\psi\rangle \otimes |\phi\rangle$, and *entangled* otherwise. Mixed-state separability is a convex-decomposition condition, not this factorization test: a mixed bipartite state is *separable* when $\rho = \sum_k r_k \, \rho_{A,k} \otimes \rho_{B,k}$ with $r_k \geq 0$ and $\sum_k r_k = 1$, and entangled otherwise. For example $\rho = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$ is correlated and not one product density operator, but its displayed convex product decomposition makes it separable.

The Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is entangled: no choice of single-qubit states $|\psi\rangle, |\phi\rangle$ gives $|\psi\rangle \otimes |\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Operators on the composite are built as $A \otimes B$ (act with A on the first factor, B on the second), though—crucially—not every operator is of this form, which is exactly how entangling gates generate entanglement from product inputs. Local products cannot entangle, but non-product form is only necessary, not sufficient, for entangling power: the SWAP operator is not $A \otimes B$ yet sends every product $a \otimes b$ to the product $b \otimes a$.

Example 12.34 (The Bell state is entangled). Suppose $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$. Matching coefficients forces $ad = bc = 0$ yet $ac = bd = \frac{1}{\sqrt{2}} \neq 0$, a contradiction. No product factorization exists—the state is entangled. ◁

Example 12.35 (A state that does factor). By contrast $\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle) = |0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |0\rangle \otimes |+\rangle$: its coefficient grid $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1 & 1\\ 0 & 0\end{smallmatrix}\right)$ has rank one, so it is a product state, not entangled. ◁

Example 12.36 (A decomposable tensor's grid). The product $(|0\rangle + |1\rangle) \otimes (|0\rangle + 2|1\rangle)$ expands to $|00\rangle + 2|01\rangle + |10\rangle + 2|11\rangle$, with coefficient grid

$$\begin{pmatrix}1 & 2\\ 1 & 2\end{pmatrix} = \begin{pmatrix}1\\ 1\end{pmatrix}\begin{pmatrix}1 & 2\end{pmatrix},$$

manifestly rank one—the algebraic signature of a product state. ◁

Example 12.37 (Creating entanglement with a gate). A Hadamard on the first qubit of $|00\rangle$ gives the product state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$; the controlled-NOT $|1x\rangle \mapsto |1\bar{x}\rangle$ then yields $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. A product input has become the entangled Bell state. ◁

Example 12.38 (The CNOT matrix). In the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ the controlled-NOT is

$$U_{\mathrm{CN}} = \begin{pmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & 1 & 0\end{pmatrix},$$

the identity on the first two basis states and a swap on the last two. It is *not* of the form $A \otimes B$; the preceding explicit product input and Bell output, not non-product form alone, proves that it can entangle. ◁

Example 12.39 (Bridge: singlet and triplet spin factors). Two spin-$\frac{1}{2}$ particles combine in $\mathbb{C}^2 \otimes \mathbb{C}^2$ into the antisymmetric *singlet* $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$ and three symmetric *triplet* states. The singlet is maximally entangled. For identical particles this spin symmetry must be combined with the spatial and other one-particle degrees of freedom before imposing symmetry on the total state. ◁

Extension: quantum-information vista. The remaining Bell-basis, GHZ, reduced-state, Schmidt, teleportation, and no-cloning material motivates later study but is not required for core progress.

Example 12.40 (Three-qubit entanglement). The GHZ state $\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ lives in $(\mathbb{C}^2)^{\otimes 3} = \mathbb{C}^8$ and is genuinely three-party entangled: no splitting into a single qubit and a pair reproduces it. Composite systems pile up one tensor factor per subsystem. ◁

Example 12.41 (A correlation). On the Bell state, the joint observable $\sigma_z \otimes \sigma_z$ has expectation $+1$—measuring σ_z on both qubits always gives equal outcomes—while the local expectations $\sigma_z \otimes I$ and $I \otimes \sigma_z$ are each 0. Perfect correlation with no local information: entanglement in a number. ◁

Example 12.42 (Correlation in the x-basis). On $|\Phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, the flip-both operator $\sigma_x \otimes \sigma_x$ of Example 12.8 gives

$$(\sigma_x \otimes \sigma_x)|\Phi\rangle = \frac{1}{\sqrt{2}}(|11\rangle + |00\rangle) = |\Phi\rangle,$$

so $\langle\Phi|\, \sigma_x \otimes \sigma_x \,|\Phi\rangle = 1$: the Bell state is perfectly correlated in the x-basis as well as the z-basis. ◁

Remark 12.43 (Extension: the Bell basis). The four Bell states $\frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$ and $\frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$ form an orthonormal basis of $\mathbb{C}^2 \otimes \mathbb{C}^2$ made entirely of maximally entangled states. A measurement in this basis—a *Bell measurement*—is the key step of teleportation and superdense coding.

Example 12.44 (Extension: the four Bell states). The Bell basis of $\mathbb{C}^2 \otimes \mathbb{C}^2$ is

$$\begin{aligned} |\Phi^\pm\rangle &= \tfrac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \\ |\Psi^\pm\rangle &= \tfrac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle), \end{aligned}$$

four orthonormal, maximally entangled states. Every two-qubit state expands in this basis, and a Bell measurement projects onto one of them. ◁

Remark 12.45 (Extension: teleportation and no cloning). A shared Bell pair lets one party send an unknown qubit using only a Bell measurement and two classical bits—entanglement plus the tensor-product structure moves quantum information without moving the particle (and not faster than light, since the classical bits are needed). Conversely, no linear U can satisfy $U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$ for all $|\psi\rangle$: unknown states cannot be cloned, a structural fact about operators on $\mathcal{H} \otimes \mathcal{H}$.

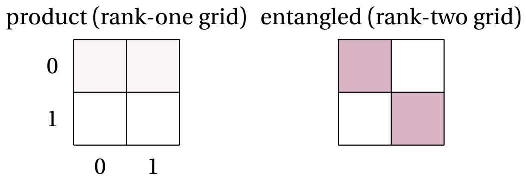

Figure 3: Coefficient grids τ^{ij} of a two-qubit state. A product state's grid is rank one (an outer product of single-qubit amplitudes); the Bell state's grid is rank two and cannot be factored—the visual signature of entanglement.

Remark 12.46 (Extension: reduced states and the partial trace). A subsystem of an entangled state has no pure state of its own; it is described by a *reduced density matrix*, the *partial trace* over the other factor (a contraction on one tensor slot). For a Bell state each qubit's reduced state is the maximally mixed $\frac{1}{2}I$—maximal ignorance of the part despite full knowledge of the whole, the hallmark of entanglement.

Example 12.47 (Extension: a reduced density matrix). For the Bell state $|\Phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, tracing out the second qubit gives $\rho_A = \mathrm{tr}_B|\Phi\rangle\langle\Phi| = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2}I$ on the first qubit. The part is maximally mixed even though the whole is a definite pure state—the defining tension of entanglement. ◁

Remark 12.48 (Extension: Schmidt decomposition). Every pure state of $\mathcal{H}_A \otimes \mathcal{H}_B$ can be written $\sum_k \lambda_k \, |a_k\rangle \otimes |b_k\rangle$ with orthonormal $|a_k\rangle$, $|b_k\rangle$ and $\lambda_k \geq 0$ (the *Schmidt decomposition*, the tensor-product cousin of the SVD). The count of nonzero λ_k is the Schmidt rank: 1 for product states, larger for entangled ones.

Remark 12.49 (Extension: exponential state-space dimension). An n-qubit system has state space $(\mathbb{C}^2)^{\otimes n} = \mathbb{C}^{2^n}$: the dimension doubles with each qubit. This exponential growth is both the promise of quantum computing—a state holds 2^n amplitudes—and the obstacle to simulating quantum systems on classical hardware. This is a dimension/amplitude count, not 2^n independently observable classical data.

Summary of main concepts

The tensor product $V \otimes W$ has basis $\{e_i \otimes f_j\}$ and dimension $(\dim V)(\dim W)$; its elements are sums of elementary tensors $v \otimes w$, most of which do not factor, and the canonical bilinear map is characterized by the universal factorization property. In coordinates this is the Kronecker product, with $(A \otimes B)(v \otimes w) = (Av) \otimes (Bw)$, $\operatorname{tr}(A \otimes B) = \operatorname{tr} A \operatorname{tr} B$, and eigenvalues $\lambda\mu$ over $\mathbb{C}$ (or after complexification). In finite dimension, a type-(p, q) tensor is a multilinear map with p upper and q lower indices, unifying vectors, covectors, the metric, and operators ($\mathcal{L}(V, W) \cong W \otimes V^*$); a real metric is bilinear, while a complex inner product is Hermitian sesquilinear. Contraction over a matched pair generalizes the trace. As a bridge, symmetric and antisymmetric tensors split $V \otimes V$ and model total exchange symmetry. In quantum mechanics finite composite pure-state vectors live in $\mathcal{H}_A \otimes \mathcal{H}_B$, and a nonzero pure state is entangled exactly when its ket does not factor. General Hilbert composites use the completed tensor product; mixed separability means a convex combination of product density operators.

- **Tensor product** $V \otimes W$—the universal recipient of bilinear maps; basis $\{e_i \otimes f_j\}$ and dimension $(\dim V)(\dim W)$.
- **Kronecker product**—the coordinate form; $\operatorname{tr}(A \otimes B) = \operatorname{tr} A \operatorname{tr} B$, complex eigenvalues $\lambda\mu$, size $nm \times nm$.
- **Finite type-(p, q) tensor**—a multilinear map with p upper and q lower indices; operators satisfy $\mathcal{L}(V, W) \cong W \otimes V^*$ and contraction generalizes the trace.
- **Bridge: symmetric and antisymmetric tensors**—split $V \otimes V$ and model total exchange symmetry.
- **Composite systems and entanglement**—finite pure-state vectors use $\mathcal{H}_A \otimes \mathcal{H}_B$; nonzero pure states are entangled iff they do not factor; mixed separability is convex.

Exercises for this lecture are in the companion sheet `exercises-12.pdf`.